Rimpei Chiba

Project Assistant Professor, National Astronomical Observatory of Japan
2-21-1 Osawa, Mitaka, Tokyo, 181-0015, Japan
rimpei.chiba@nao.ac.jp
ORCID: 0000-0002-3445-855X

EMPLOYMENT

2026 - 2031 **Project Assistant Professor**, National Astronomical Observatory of Japan
2025 - 2026 **JSPS Postdoctoral Fellow**, University of Tokyo
2022 - 2025 **CITA Postdoctoral Fellow**, University of Toronto

EDUCATION

2018 - 2022 *D.Phil. in Theoretical Physics*, University of Oxford
2016 - 2018 *M.Sc. in Physics*, Tokyo Institute of Technology & TU Wien
2012 - 2016 *B.Sc. in Aerospace Engineering*, Tokyo Institute of Technology

PUBLICATION (refereed)

Leading authors

[9] **Chiba R.**, Fujii M., Baba J., Dubinski J., Schönrich R., "Tree-ring structure of Galactic bar resonance in N-body simulations", ApJ, 1006, 132-148 (2026)

[8] **Chiba R.**, Ding J., Hamilton C., Kunz M. W., Tremaine S., "Galactic echoes", MNRAS, 543, 190-201 (2025)

[7] **Chiba R.**, Frankel N., Hamilton C., "Origin of the two-armed vertical phase-spiral in the inner Galactic disk", MNRAS, 543, 2159-2179 (2025)

[6] **Chiba R.**, Kataria S. K., "Origin of reduced dynamical friction by dark matter halos with net prograde rotation", MNRAS, 528, 4115-4124 (2024)

[5] **Chiba R.**, "Dynamical friction and feedback on galactic bars in the general fast-slow regime", MNRAS, 525, 3576-3596 (2023)

[4] **Chiba R.**, Schönrich R., "Oscillating dynamical friction on galactic bars by trapped dark matter", MNRAS, 513, 768-787 (2022)

[3] **Chiba R.**, Schönrich R., "Tree-ring structure of Galactic bar resonance", MNRAS, 505, 2412-2426 (2021)

[2] **Chiba R.**, Friske J., Schönrich R., "Resonance sweeping by a decelerating Galactic bar", MNRAS, 500, 4710-4729 (2021)

[1] **Chiba R.**, Ishikawa Y., Hasegawa J., Horioka K., "Time evolution of laser-ablation plumes and induced shock waves in low-pressure gas", AIP, Physics of Plasmas, 24, 063520 (2017)

Co-authors

[7] Frankel N., Semczuk M., Antoja T., Chakrabarti S., **Chiba R.**, Grand R. J. J., Hunt J. A. S., Khoperskov S., Li Z.-Y., Lutsenko A., Ramos P., Tavangar K., Widrow L. M., "Winding, Unwinding, Rewinding the Gaia Phase Spiral", PASP, 138, 066001, (2026)

[6] Kodama R., **Chiba R.**, Asano T., Baba J., Fujii M., "Galaxy fly-bys sustain bar–halo friction and bar slowdown in disk galaxies", PASJ, 78(3), 947–957, (2026)

[5] Géron T., Smethurst R. J., Dickinson H., Fortson L. F., Garland I. L., Kruk S., Lintott C., Makechemu J. S., Mantha K. B., Masters K. L., O'Ryan D., Roberts H., Simmons B. D., Walmsley M., Calabrò A., **Chiba R.**, Costantin L., Drout M. R., Fragkoudi F., Guo Y., Holwerda B. W., Jogee S., Koekemoer A. M., Lucas R. A., Pacucci F., "Galaxy Zoo CEERS: Bar fractions up to $z \sim 4.0$", ApJ, 987:74, (2025)

[4] Li C., Yuan Z., Monari G., Martin N. F., Siebert A., Famaey B., **Chiba R.**, Kordopatis G., Ibata R. A., Hill V., "Exploring the impact of a decelerating bar on transforming bulge orbits into disc-like orbits", A&A, Vol 690, A26 (2024)

[3] Yuan Z., Li C., Martin N. F., Monari G., Famaey B., Siebert A., **Chiba R.**, Ardern-Arentsen A., Sestito F., Thomas G. F.,7, Hill V., Ibata R. A., Kordopatisc G., Starkenburg E., Viswanathan A., "Could very low-metallicity stars with rotation-dominated orbits have been driven by the bar?", A&A, Vol 691, L1 (2024)

[2] Szabo P., **Chiba R.**, Biber H., Stadlmayr R., Berger B., Mayer D., Mutzke A., Doppler M., Sauer M., Appenroth J., Fleig J., Foelske-Schmitz A., Hutter H., Mezger K., Lammer H., Galli A., Wurz P., Aumayr F., "Solar wind sputtering of wollastonite as a lunar analogue material – Comparisons between experiments and simulations", Icarus, Vol 314, 98-105 (2018)

[1] Stadlmayr R., Szabo P., Berger B., Cupaka C., **Chiba R.**, Blöcha D., Mayer D., Stechauner B., Sauer M., Foelske-Schmitz A., Oberkofler M., Schwarz-Selinger T., Mutzke A., Aumayr F., "Fluence dependent changes of surface morphology and sputtering yield of iron: Comparison of experiments with SDTrimSP-2D", Nuclear Inst. and Methods in Physics Research B, Vol 430, 42-46 (2018)

SCHOLARSHIPS AND AWARDS

2025 - 2028	**JSPS Postdoctoral Fellowship**
2022 - 2025	**CITA Postdoctoral Fellowship**
2018 - 2022	**Takenaka Scholarship** for overseas studies
2018	**Best Master Thesis Award** (Tokyo Institute of Technology)
2017	**Young Poster Award** (International Astronomical Union, Symposium 335)
2016 - 2017	**Tobitate Scholarship** (Japan Young Ambassador Program)

SCIENTIFIC PRESENTATION

6/2026	Contributed Talk, Galactic Bars, Bulges and Disks Workshop (Lipari, Italy)
4/2026	Invited Talk, Galactic center workshop (NAOJ, Japan)
3/2026	Contributed Talk, Japan Astronomical Society Annual meeting (Kyoto, Japan)
11/2025	Invited Talk, Milky Way workshop (Kagoshima, Japan)
10/2025	Talk, Tohoku University, Astronomy Colloquium (Sendai, Japan)
09/2025	Contributed Talk, Japan Astronomical Society Annual meeting (Shimonoseki, Japan)
08/2025	Invited Review Talk, Workshop "Winding, Unwinding and Rewinding the Gaia Phase Spiral" (Leiden, Netherland)
02/2025	Talk, Como Lake centre for AstroPhysics, Colloquium (Como, Italy)

01/2025	Invited Talk, Conference IAStrophysical Waves (Princeton, US)
09/2024	Talk, Nanjing University, Astrophysics seminar (Nanjing, China)
09/2024	Invited Talk, Conference "Waves in the Milky Way Disk" (Shanghai, China)
06/2024	Invited Blackboard Talk, KITP Program "Interconnections between the Physics of Plasmas and Self-gravitating Systems" (Santa Barbara, US)
01/2024	Talk, Shanghai Jiao Tong University, Astronomy seminar (Shanghai, China)
09/2023	Talk, Institute for Advanced Study, Astrophysics seminar (Princeton, US)
07/2023	Poster, Galactic bars: driving and decoding galaxy evolution (Granada, Spain)
06/2023	Contributed Talk, Canadian Astronomical Society, Annual meeting (Penticton, Canada)
12/2022	Talk, CITA seminar (Toronto, Canada)
11/2022	Talk, University of Leicester, Astrophysics seminar (Leicester, UK)
01/2022	Talk, University College London, MSSL seminar (London, UK)
07/2021	Contributed Talk, National Astronomy Meeting (Bath, UK)
02/2021	Talk, University of Oxford, Galaxy Journal Club seminar (Oxford, UK)
07/2020	Talk, European Astronomical Society, Annual meeting (Leiden, Sweden)
12/2019	Talk, University of Surrey, Milky Way Local Group seminar, (Surrey, UK)
09/2017	Contributed Talk, The Physical Society of Japan, Annual meeting (Iwate, Japan)
07/2017	Poster, International Astronomical Union, Symposium 335 (Exeter, UK)
04/2017	Poster, European Geoscience Union, General Assembly (Vienna, Austria)
01/2016	Contributed Talk, National Institute of Fusion Science, Conference (Gifu, Japan)

ACADEMIC SERVICES

2022 - 2025	CITA blackboard seminar, organizer
Review:	Monthly Notices of the Royal Astronomical Society, Astronomy & Astrophysics, Astrophysical Journal

SKILLS

Computing:	C/C++, python, MPI, OpenMP
Language:	Japanese (native), English (fluent)

REFERENCES

- **Ralph Schönrich**
  Mullard Space Science Laboratory, University College London
  r.schoenrich@ucl.ac.uk
- **John Magorrian**
  Rudolf Peierls Centre for Theoretical Physics, University of Oxford
  john.magorrian@physics.ox.ac.uk
- **James Binney**
  Rudolf Peierls Centre for Theoretical Physics, University of Oxford
  james.binney@physics.ox.ac.uk
- **Peter Martin**
  Canadian Institute for Theoretical Astrophysics, University of Toronto
  pgmartin@cita.utoronto.ca
- **Scott Tremaine**
  School of Natural Sciences, Institute for Advanced Study
  tremaine@ias.edu

• **Michiko Fujii**
Department of Astronomy, University of Tokyo
fujii@astron.s.u-tokyo.ac.jp